

FE Front MGU — Jack verdict (one page)

Concept pack under named assumptions · not homologation · ship_ok = false

Architecture freeze (DEC-009): 24000.0 rpm · 130.0 mm stack · magnets 6.0×22.5 mm
Duty assumption (DEC-008): intermittent peak (vignette ~24 s regen / 100 s) – not continuous 250 kW thermal

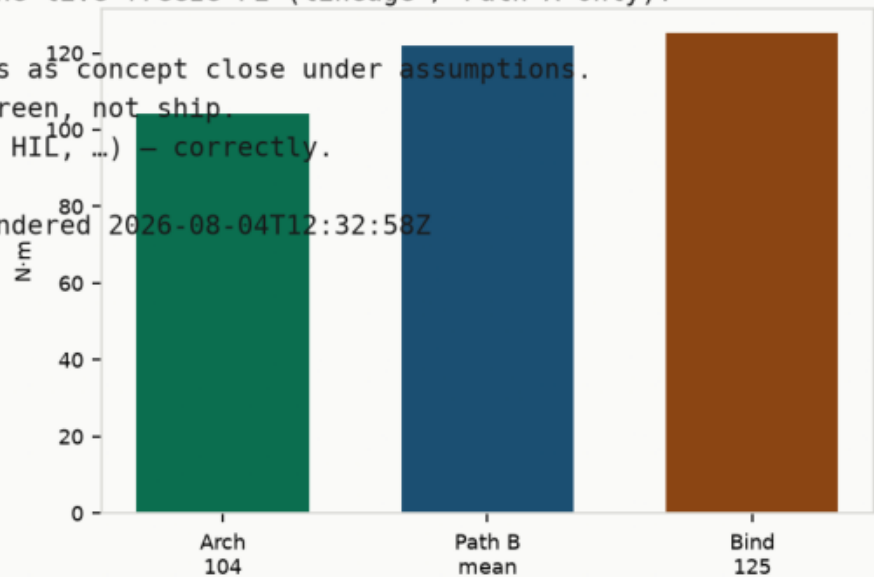
Path B kit-case FE mean T	122.100 N·m	(sign-consistent; sign_reversals=0; SIGHT-candidate only)
Architecture duty bar (24k)	104.099 N·m	→ mean/arch = 1.173× gap = +18.0 N·m CLEARS
Conservative binding (ledger)	125.215 N·m	→ mean/bind = 0.975× gap = -3.1 N·m DOES NOT CLEAR

duty_torque_screen_ok = False torque_reliable = False
Fail reason is the reliability gate (dyno/map), not short torque vs architecture bar.

Product field mgu_fe_shaft_torque_nm remains option-screen PRODUCT (not kit-case FE).
Do not quote failed DEC009 artefacts (re-derived magnets 8.85×14.58 mm).
Do not quote pre-DEC-009 REBALANCED mean 81.56 N·m as the live freeze FE (lineage / Path A only).

Bar A process: A-DUTY re-freeze under DEC-008/009 stands as concept close under assumptions.
Bar A FE: Path B is SIGHT-candidate – not duty-screen green, not ship.
Bar B: B1–B10 all OPEN (dyno, Gerbers, XYZ, flow bench, HIL, ...) – correctly.

Artefact: em_fia_front_kit_case_PATH_B_DEC009.json rendered 2026-08-04T12:32:58Z



Open by design — Bar B & partner holds

Software can define and ask. Software cannot close these. ship_ok stays false.

Pri	ID	Item	Class	Status
1	BARB-DUTY-CYCLE	Duty-cycle / E_net authority (DEC-007)	ASSUMED_CONCEPT	OPEN
2	BARB-DYNO	Motor + inverter dyno correlation (DEC-0	NEEDS_HARDWARE	OPEN
3	BARB-FLOW-BENCH	Flow bench jacket + cold plate	NEEDS_HARDWARE	OPEN
4	BARB-ICD-XYZ	FIA / chassis port XYZ	NEEDS_PARTNER_IN	OPEN
5	BARB-GERBERS	Supplier Gerbers / pinout ICD (DEC-009)	NEEDS_PARTNER_IN	OPEN
6	BARB-SIC-MODULE	SiC module identity + thermal limit (DEC	ASSUMED_CONCEPT	OPEN
7	BARB-ROTOR-RETENTION	Rotor retention / overspeed (DEC-006)	ASSUMED_CONCEPT	OPEN
8	BARB-HIL	HIL on populated inverter (DEC-008)	NEEDS_HARDWARE	OPEN
9	BARB-HEATER-PLATE	Heater-plate module temperatures	NEEDS_HARDWARE	OPEN
10	BARB-DOUBLE-PULSE	Double-pulse ESL / switching	NEEDS_HARDWARE	OPEN

How to read this pack

CONCEPT (software-closed under assumptions)

- DEC-008 intermittent duty + DEC-009 24k/130
- Path B sign-stable kit-case FE mean
- Dual torque bars (architecture vs conservative)
- Cooling / oil / rotor screens (screening class)
- Draft PCB topology (NOT_FAB)
- Blender morphology + EM honesty figures

You may treat this as a serious concept FPK.

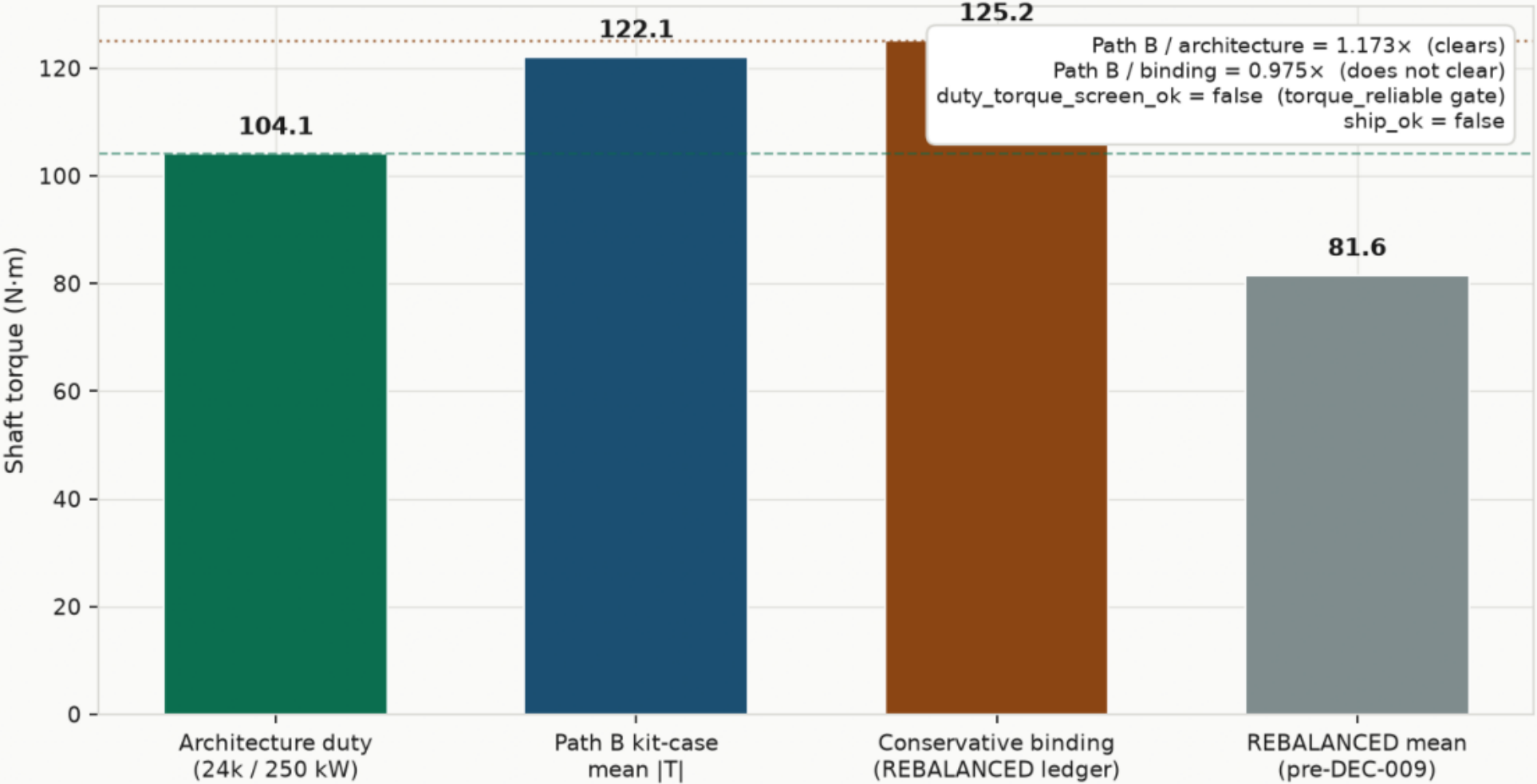
HOMOLOGATION / RACE (partner + hardware)

- Dyno map → torque_reliable
- Lap CSV → duty authority
- Supplier Gerbers + SiC MPN
- Chassis XYZ mounts
- Flow bench / oil CFD + clear case
- Release FEA + material certs
- HIL on populated inverter

ship_ok stays false until these exist.

If a figure looks green, read the caption. Green architecture bar ≠ green duty screen ≠ ship.

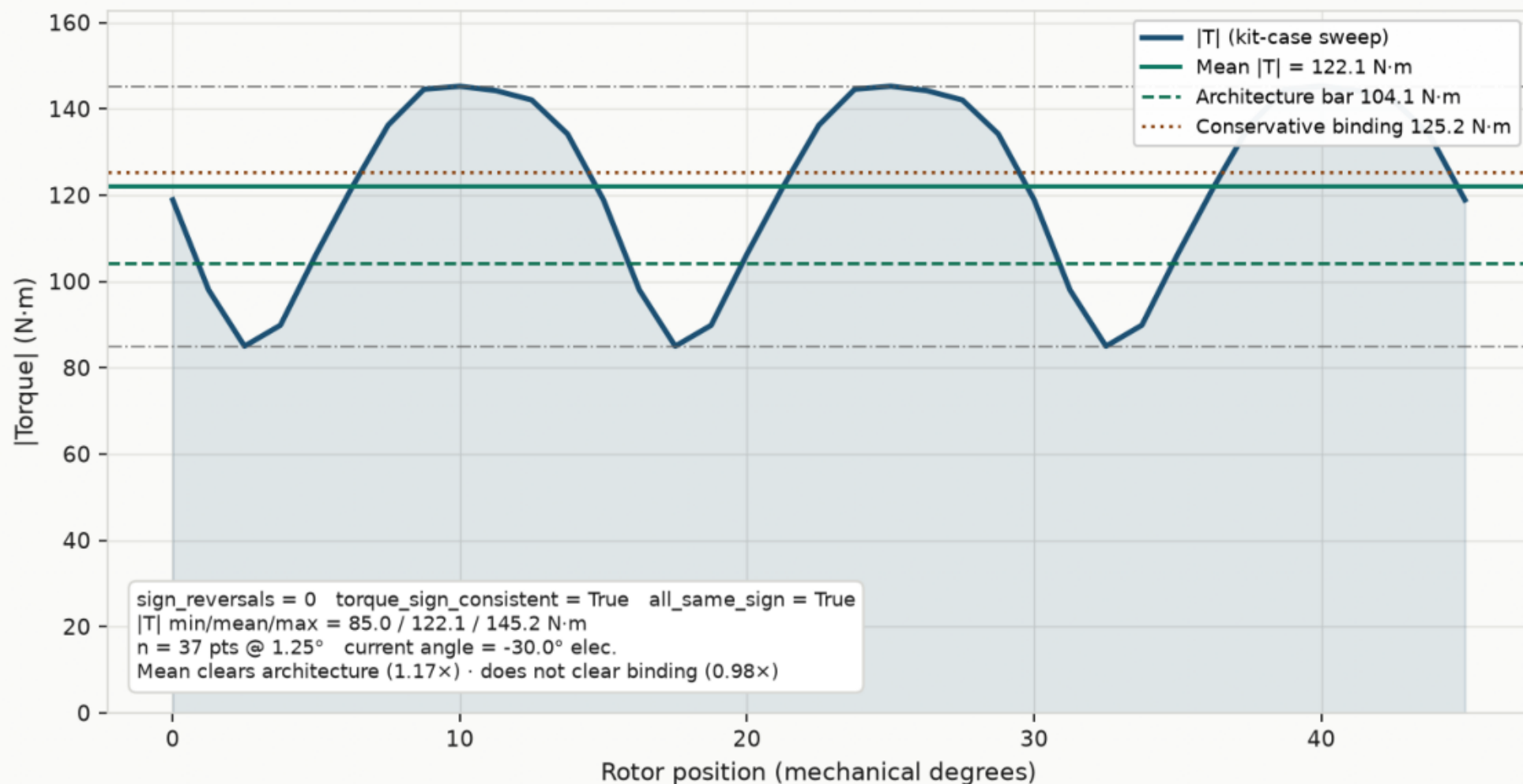
FE Front — dual torque bars (honest reading)



Architecture bar = Path B analytical required shaft torque at DEC-009 freeze (24,000 rpm, 250 kW, twin η). Conservative binding = REBALANCED ledger bar (~19.5k class). Path B mean is sign-consistent kit-case FE — not duty-clear SIGHT, not ship_ok.

Source: em_fia_front_kit_case_PATH_B_DEC009.json · rendered 2026-08-04T12:32:58Z

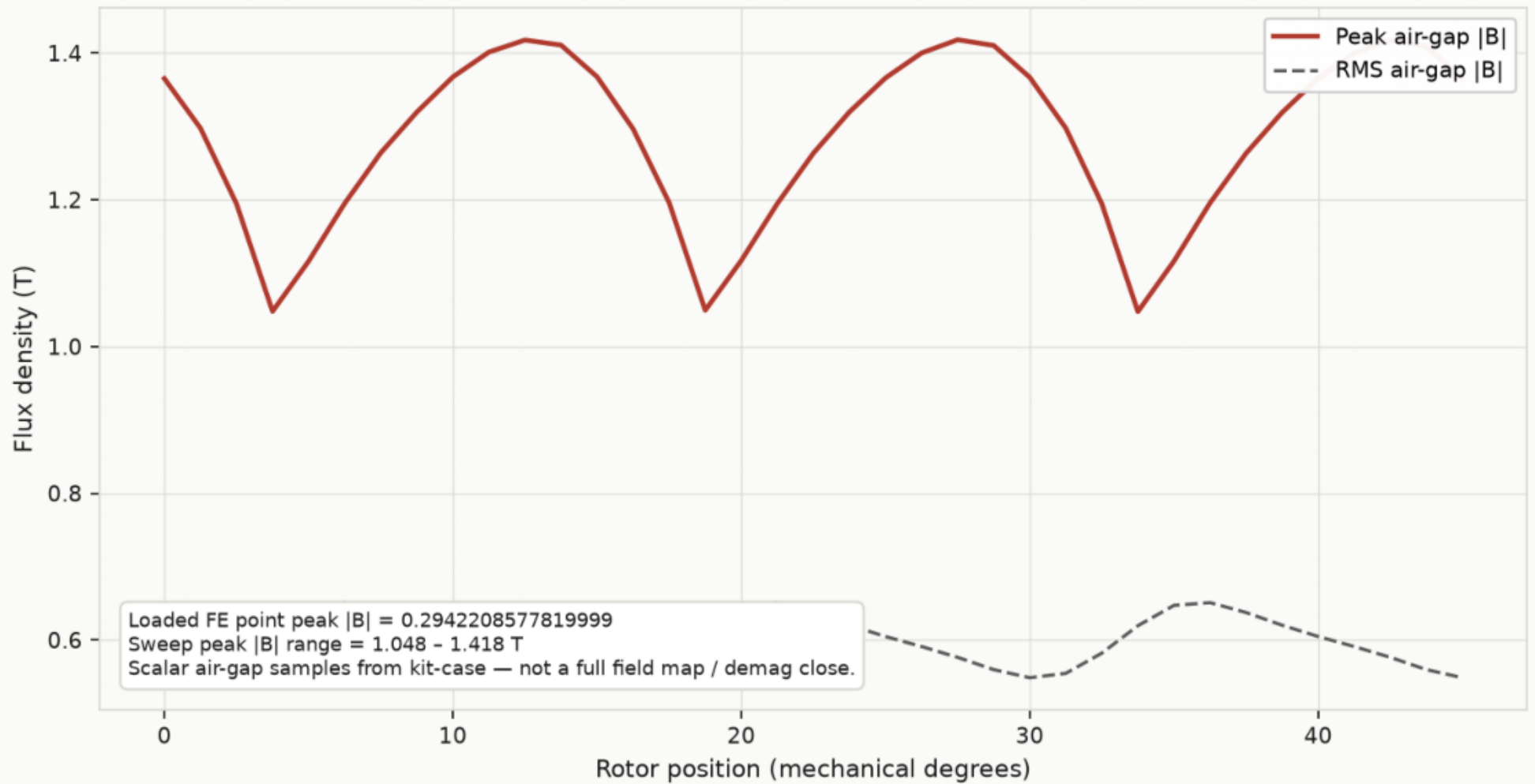
Path B — |torque| vs rotor position (fixed current angle -30° elec.)



Coarse mechanical rotor-position sweep at one screened current angle — not a full MTPA map. Zero sign reversals is the Path A/B honesty gate that failed on the old DEC009 (re-derived magnets).

Source: em_fia_front_kit_case_PATH_B_DEC009.json#rotor_position_sweep · rendered 2026-08-04T12:32:58Z

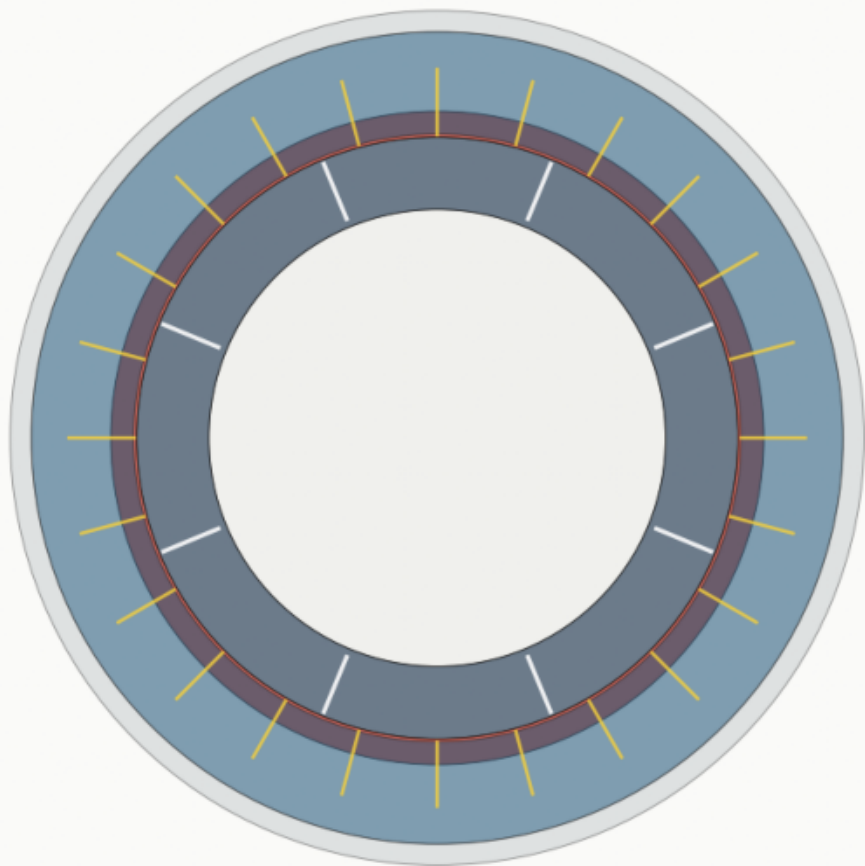
Path B — air-gap flux density vs rotor position



Physics flavour from the same Path B sweep used for torque. Does not claim MTPA, demagnetisation, or thermal closure.

Source: em_fia_front_kit_case_PATH_B_DEC009.json · rendered 2026-08-04T12:32:59Z

Path B machine identity — DEC-009 freeze (cross-section schematic)



Active length	130.00 mm
Rotor OD	139.4 mm
Stator OD	188.2 mm
Housing OD	198.0 mm
Air-gap	0.70 mm
Magnets	6.0 × 22.5 mm (baseline freeze)
Topology	24 slots / 8 poles
Speed	24000.0 rpm
I_design	535.0 A rms
Mean T	122.099939 N·m

Schematic only (not CAD). Magnets shown as radial band for identity — IPM pocket detail is in kit-case geometry.

Identity card for Path B DEC-009 freeze. Failed DEC009 used re-derived magnets 8.85×14.58 mm — do not quote that artefact.
Source: em_fia_front_kit_case_PATH_B_DEC009.json#machine_geometry · rendered 2026-08-04T12:32:59Z

FE Front MGU — Path B EM status

Architecture 24000 rpm · 130.0 mm stack

Magnets 6.0 × 22.5 mm (frozen)

Kit-case mean 122.1 N·m (sign-consistent)

Architecture bar 104.1 N·m → 1.17× CLEAR

Conservative bar 125.2 N·m → 0.98× NOT CLEAR

duty_torque_screen_ok = false

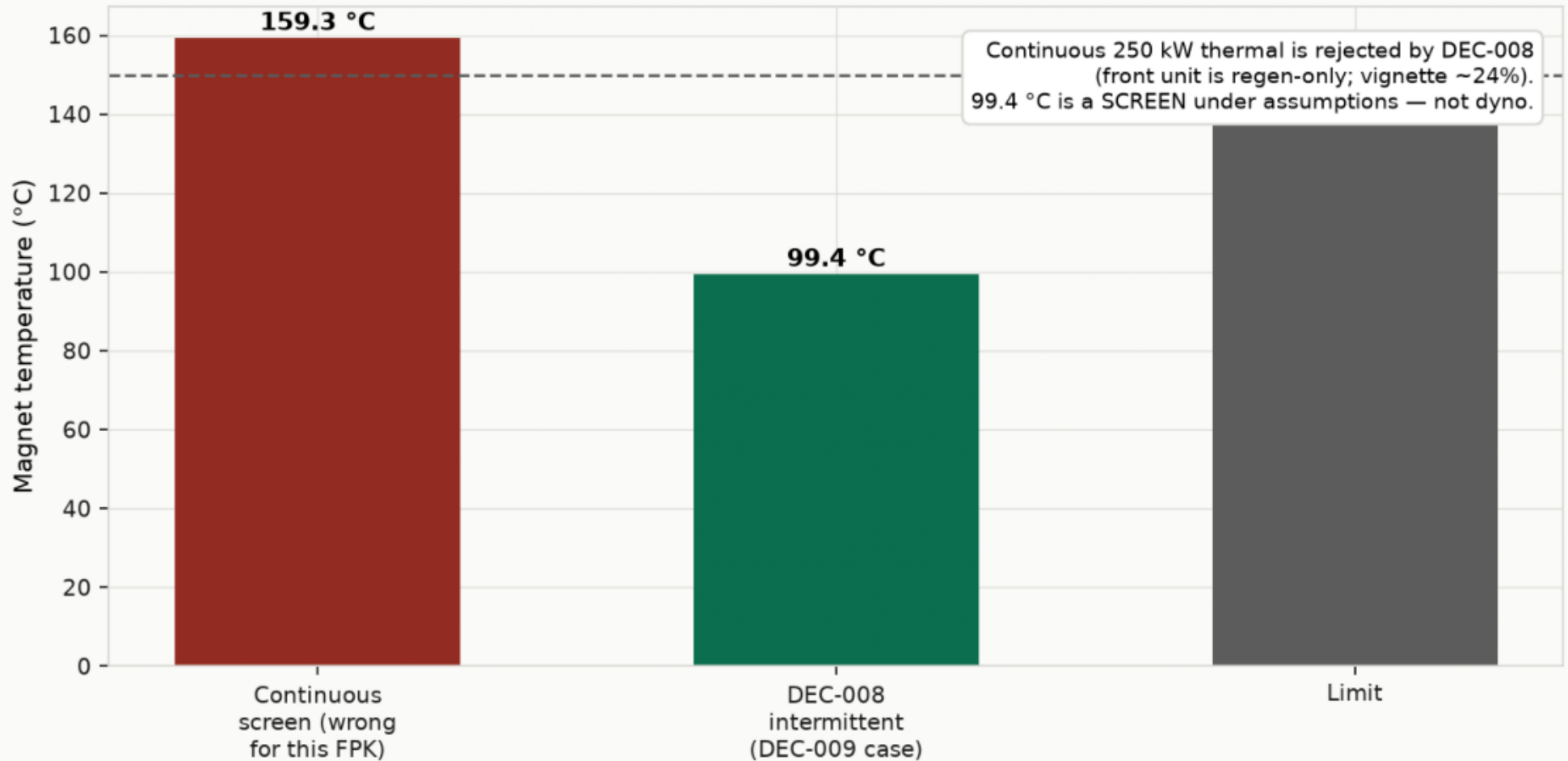
(torque_reliable gate — dyno/map still open)

ship_ok = false · not Bar A close

Do not quote failed DEC009 (8.85 mm magnets).



Thermal storyboard — magnet temperature under duty assumption



DEC-008 makes intermittent duty the design basis; continuous magnet 159 °C was an artefact of screening 24% duty at 100% load.

Sources: state quantities magnet_temperature_screen_c / mgu_magnet_temp_c · 2026-08-04T12:32:59Z

PCB honesty sheet — draft topology, not fabrication

disposition=bespoke · NOT_FABRICATION_READY · ship_ok=false · 2026-08-04T12:44:27Z

- **traction_gate_drive** (**traction_gate_drive_board**)

outline ~120×90 mm · mounts=6 · domains=['power', 'high_voltage']

channels: gate_drive_channel×6, desat_channel×6

work: isolate_gate_drive, desat_protect, drive_sic_half_bridges

- **traction_control** (**traction_control_board**)

outline ~100×80 mm · mounts=4 · domains=['logic', 'analog']

channels: phase_current_sense×3, resolver_channel×1, vehicle_can×1, lv_buck_rail×3, hv_lv_isolation_barrier×1

work: sense_phase_current, demodulate_resolver, vehicle_can, regulate_lv_rails, isolate_hv_lv_domains

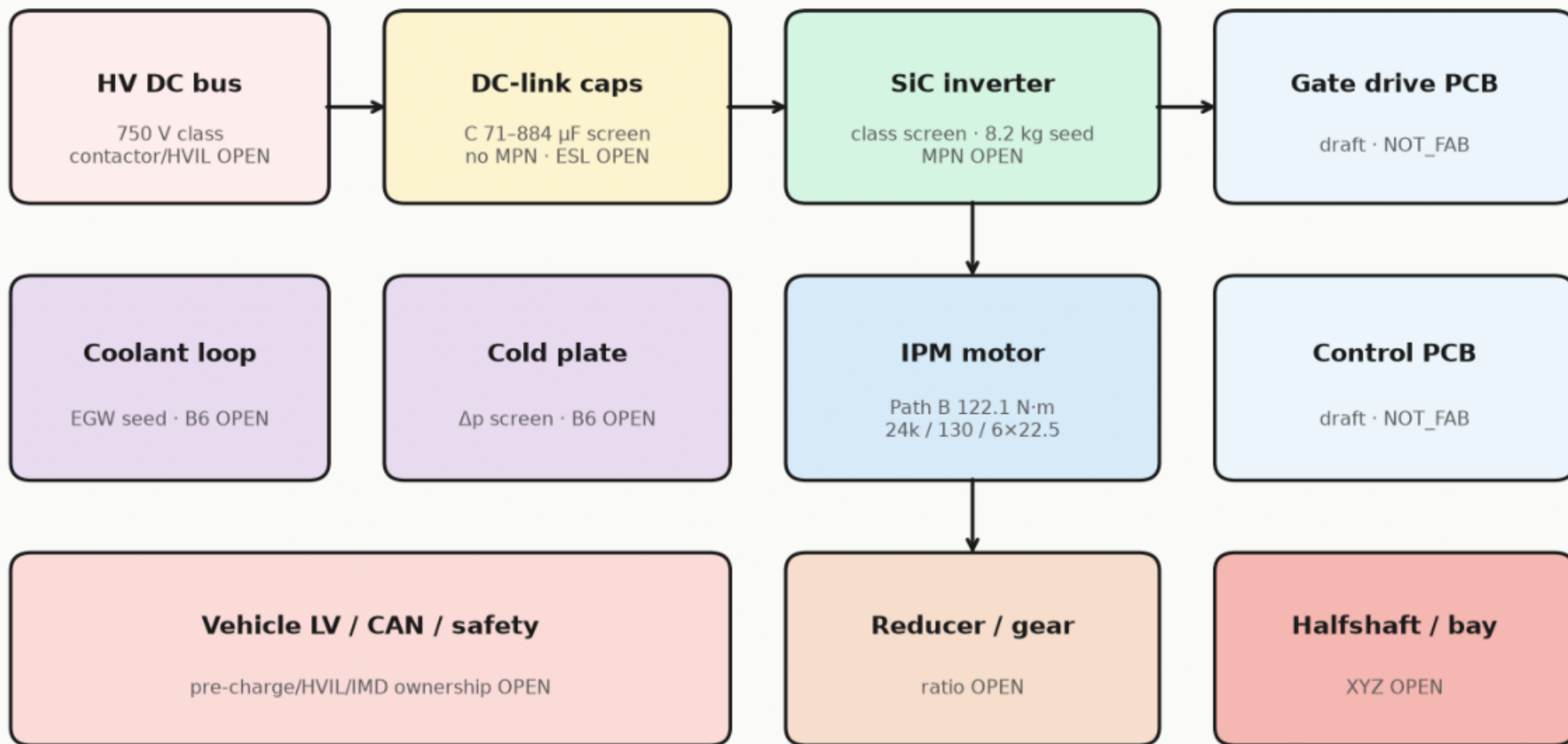
HAVE (software)

- ✓ Forge-authored KiCad drafts for gate-drive + control
- ✓ Channel architecture from twin words
- ✓ DRC pipeline on twin (forgeDraftOnly)

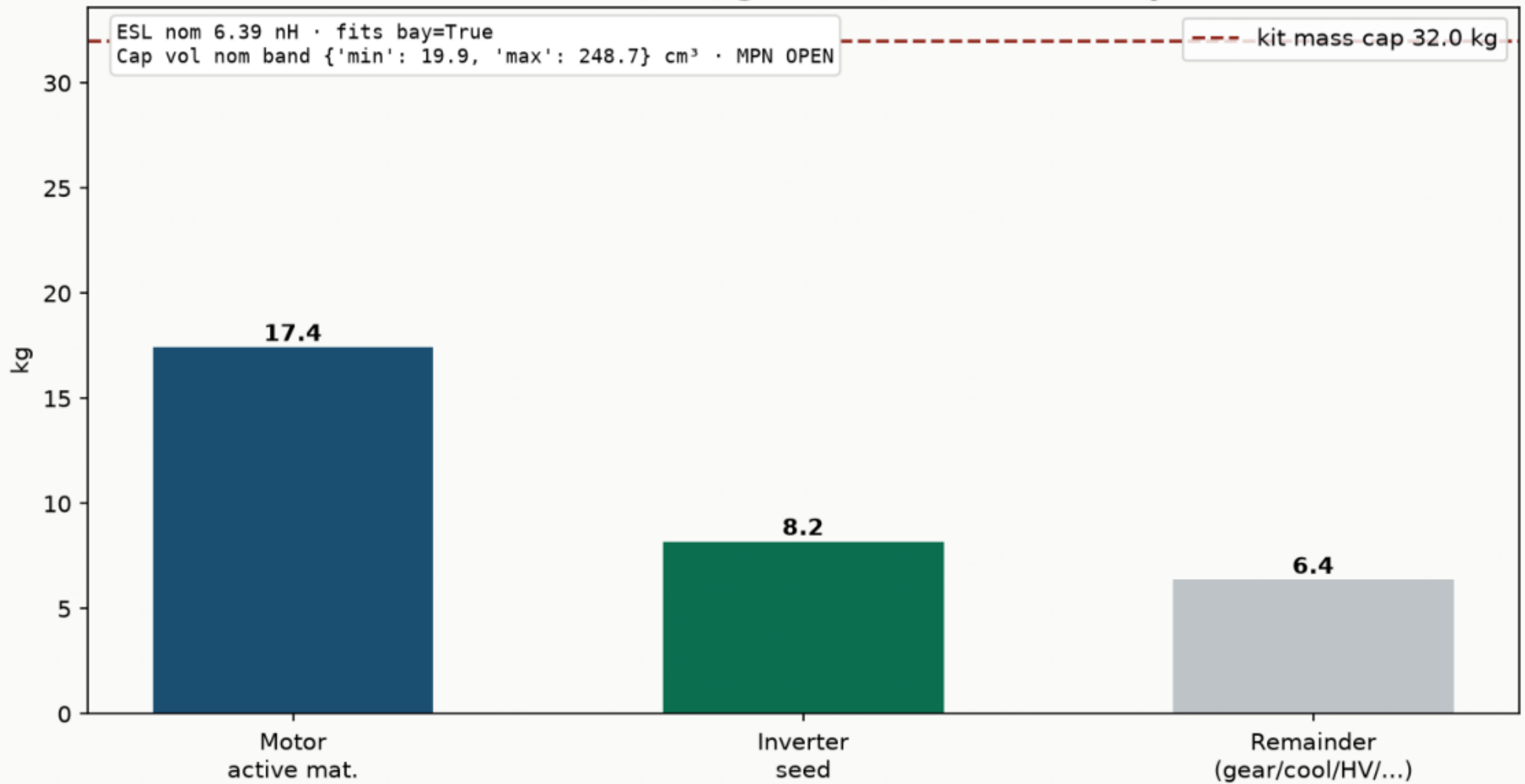
DO NOT HAVE (Bar B / partner)

- Supplier-stamped Gerbers / stack-up
- Frozen SiC module MPN pinout ICD
- HIL on populated hardware
- Production BOM / fab release

Front FPK system block — energy & control boundaries (concept)



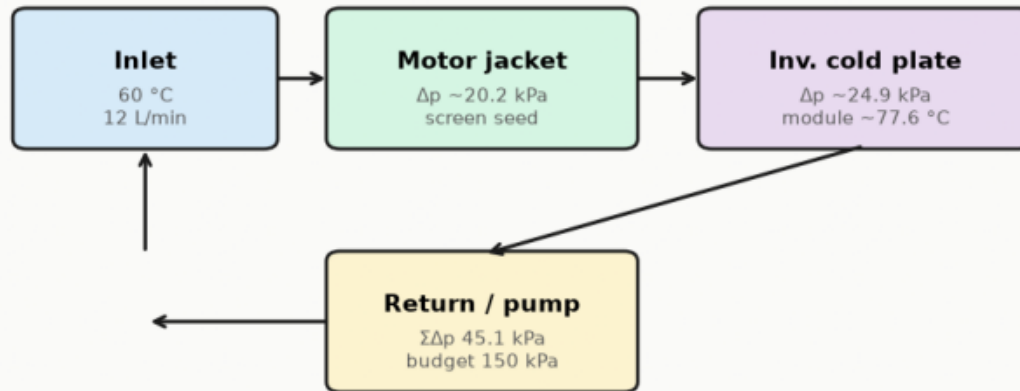
Inverter class + mass budget reconciliation (concept seeds)



Coolant loop one-pager — analytical network seeds (W2.6)

EGW class · flow_bench OPEN · ship_ok=false · 2026-08-04T13:09:09Z

Series hydraulic path (concept) — not a CFD mesh



Σ Δp **45.1 kPa**

Module T **77.6 °C**

Coolant ΔT **17.2 K**

Coupled screen **OK**

Magnet temperature honesty (twin stamps)

DEC-008 intermittent vignette ~99.4 °C · continuous screen ~159.3 °C · limit 150 °C

Continuous magnet stamp exceeds limit — duty_torque_screen stays false; intermittent vignette is the architecture path.

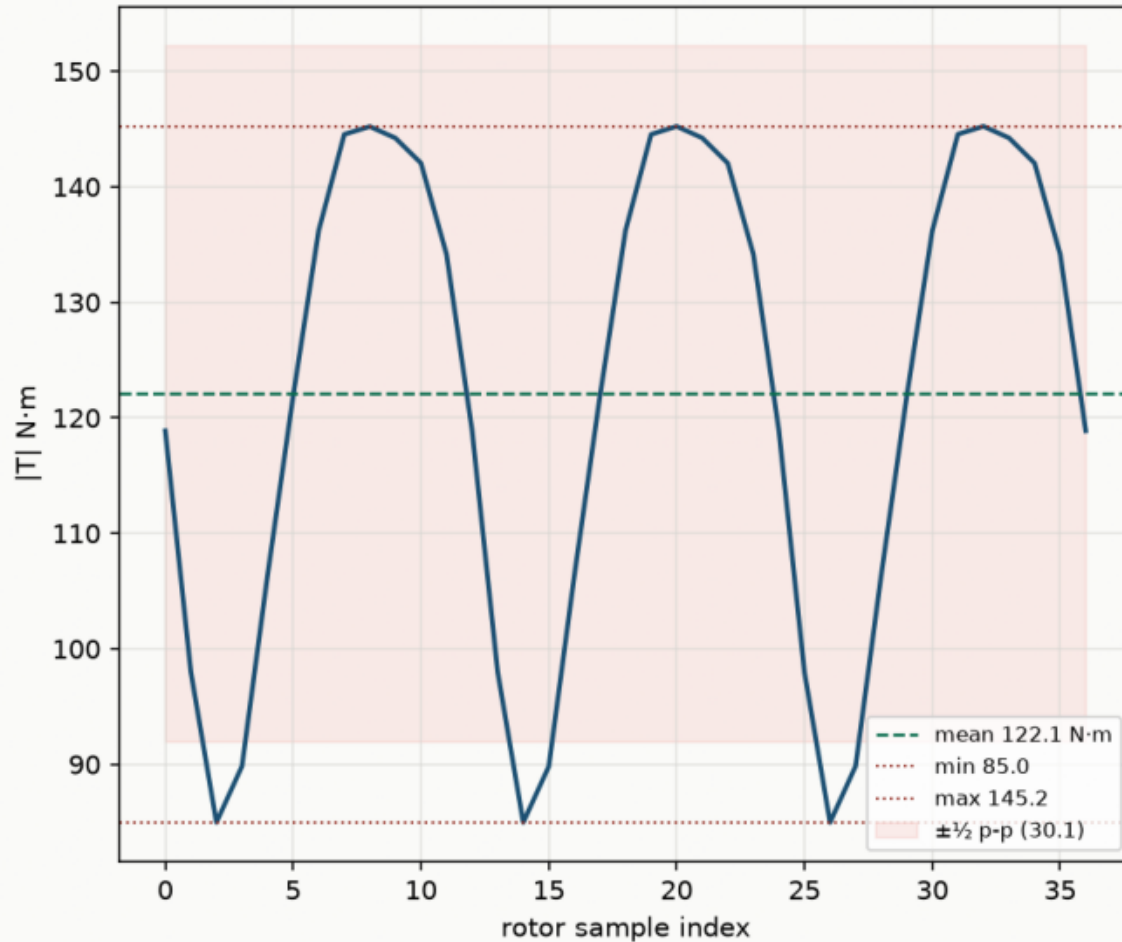
Flow bench must prove (B6 / BARB-FLOW-BENCH)

- Jacket + cold-plate Δp vs flow at 12 L/min class (and neighbour points)
- Wall tap temperatures vs network prediction at known loss inject
- Parallel branch balance (no starvation of cold plate)
- No free-stream leak paths / QD Δp contribution
- Same fluid identity, inlet T, and total-vs-per-channel flow basis as network

W2.6 · analytical network only · no measured Rth · no CHT · ship_ok=false

Torque-ripple load sheet — Path B $|T|$ sweep (W2.8)

Sign-stable magnitude sweep



min / mean / max 84.97 / 122.10 / 145.23 N·m
peak-to-peak 60.26 N·m
 $\frac{1}{2}$ -range amplitude 30.13 N·m
samples 37

rotor OD 139.4 mm
status INPUT_NOT_VALIDATED
torque_reliable False
duty_screen_ok False
F_tang mean 1.752 kN
F_tang $\frac{1}{2}$ -range 0.432 kN
def: $F=|T|/(r_{od}/2)$ (not bearing)

NOT claimed / must not use as:

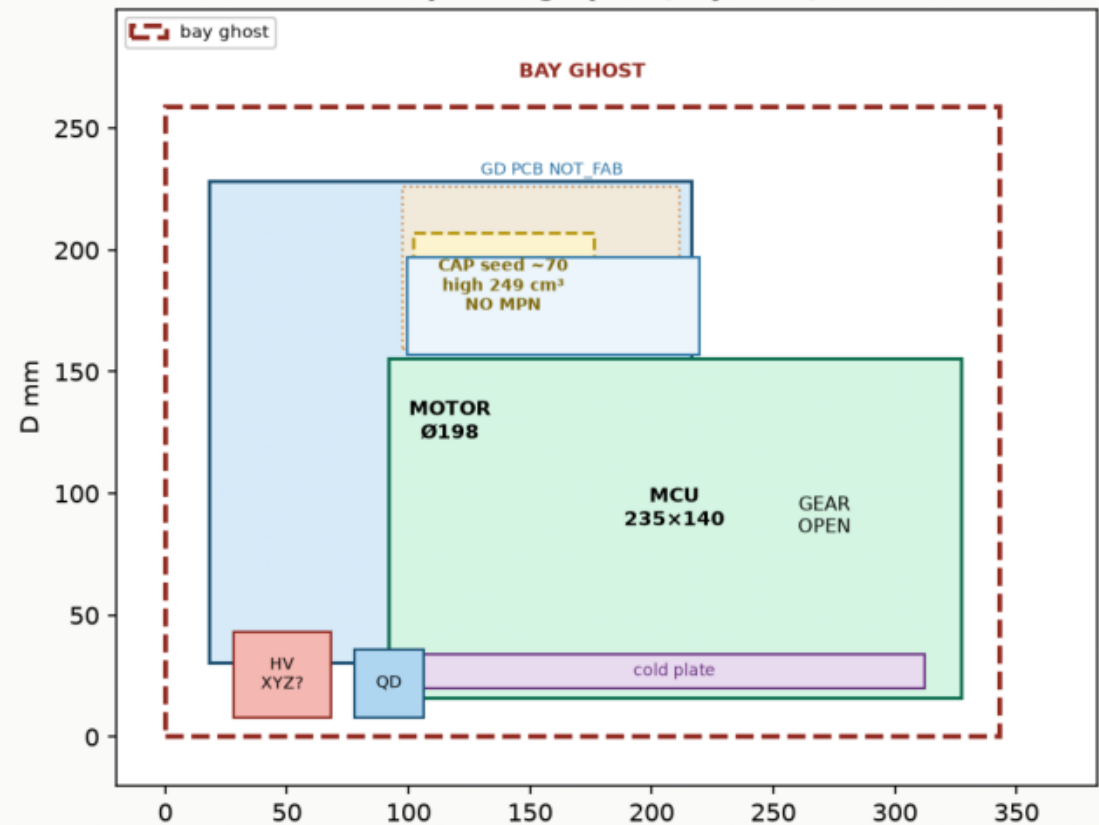
- bearing L10 / gear mesh force
- validated cyclic load
- torque_reliable / dyno
- ship_ok

Partner ask: BARB-DYNO map @ Path B OP

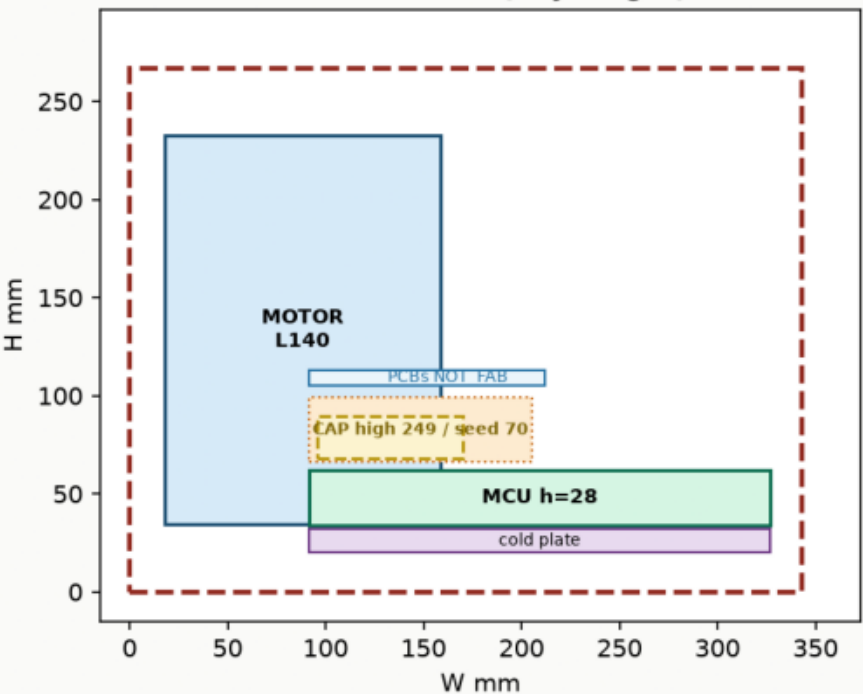
Kit assembly hierarchy — W3.2-lite envelopes (visualOnly)

Bay ghost 343×259×267 mm · no Blender hero · ship_ok=false · 2026-08-04T13:09:09Z

Top orthographic (bay floor)



Side elevation (bay height)



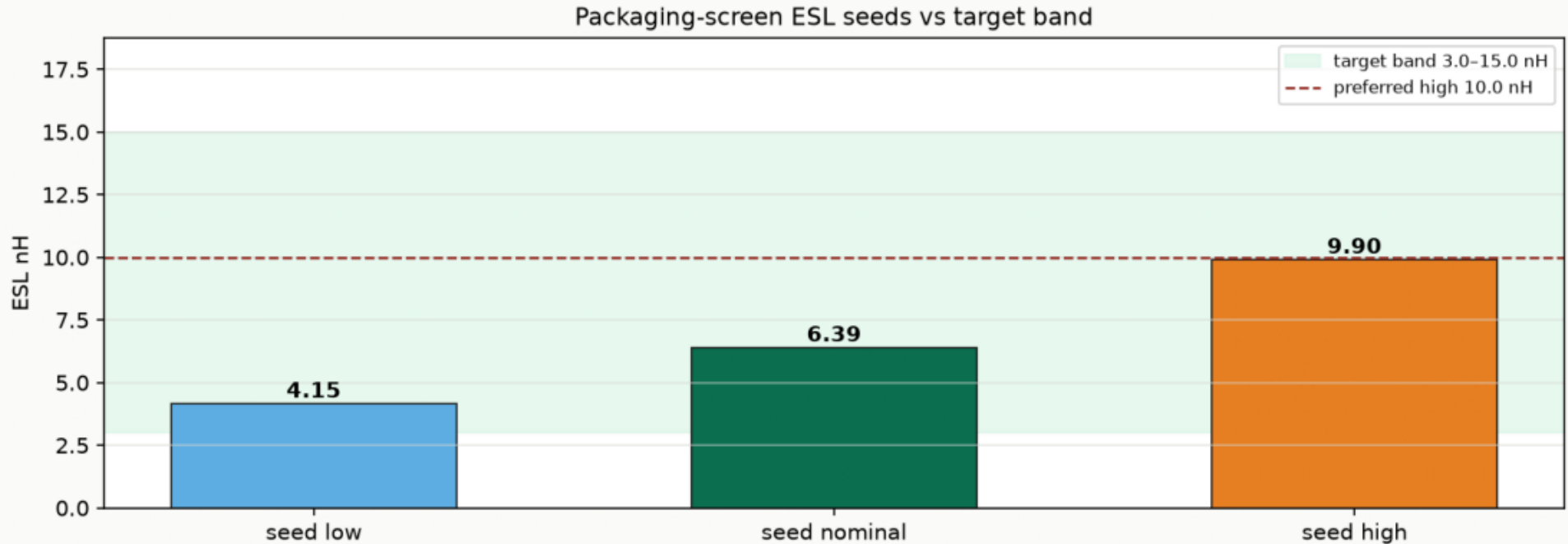
Part hierarchy (visualOnly envelopes) W mm

- IPM motor housing 198.0× 198.0×140.5 mm [PATH_B_GEOMETRY] OD×L · active L 130.0 mm
- Reducer / gear 90.0× 80.0× 70.0 mm [OPEN_RATIO] placeholder box – ratio OPEN
- Inverter MCU housing 235.3× 139.6× 28.0 mm [PACKAGING_SCREEN] fits_bay=True
- Cold plate land 235.3× 139.6× 12.0 mm [SEED_THICKNESS] h=12.0 mm seed · B6 OPEN
- DC-link capacitor region (geo-mean seed) 74.5× 43.5× 21.7 mm [CONCEPT_ENVELOPE_SEED] vol geo-mean~70 cm³ · NOT conservative fit
- DC-link capacitor HIGH volume (fit check) 113.5× 66.2× 33.1 mm [CONCEPT_ENVELOPE_CONSERVATIVE] vol high=249 cm³ under ASSUMED aspect 2.4:1.4:0.7 – not a datasheet envelope; fit is sha
- DC-link capacitor LOW volume 48.9× 28.5× 14.3 mm [CONCEPT_ENVELOPE_LOW] vol low=20 cm³

W3.2-lite · boxes from twin seeds + C volume band · XYZ mounts OPEN · not CAD release · ship_ok=false

Busbar / loop inductance budget — analytical (W2.3)

Laminated-bus ESL seeds · double-pulse OPEN · ship_ok=false · 2026-08-04T14:41:19Z



Switching spike order-of-magnitude $V \approx L \cdot di/dt$ (ASSUMED di/dt 2-8 kA/ μ s)

At nom 6.39 nH: $\Delta V \sim 13-51$ V

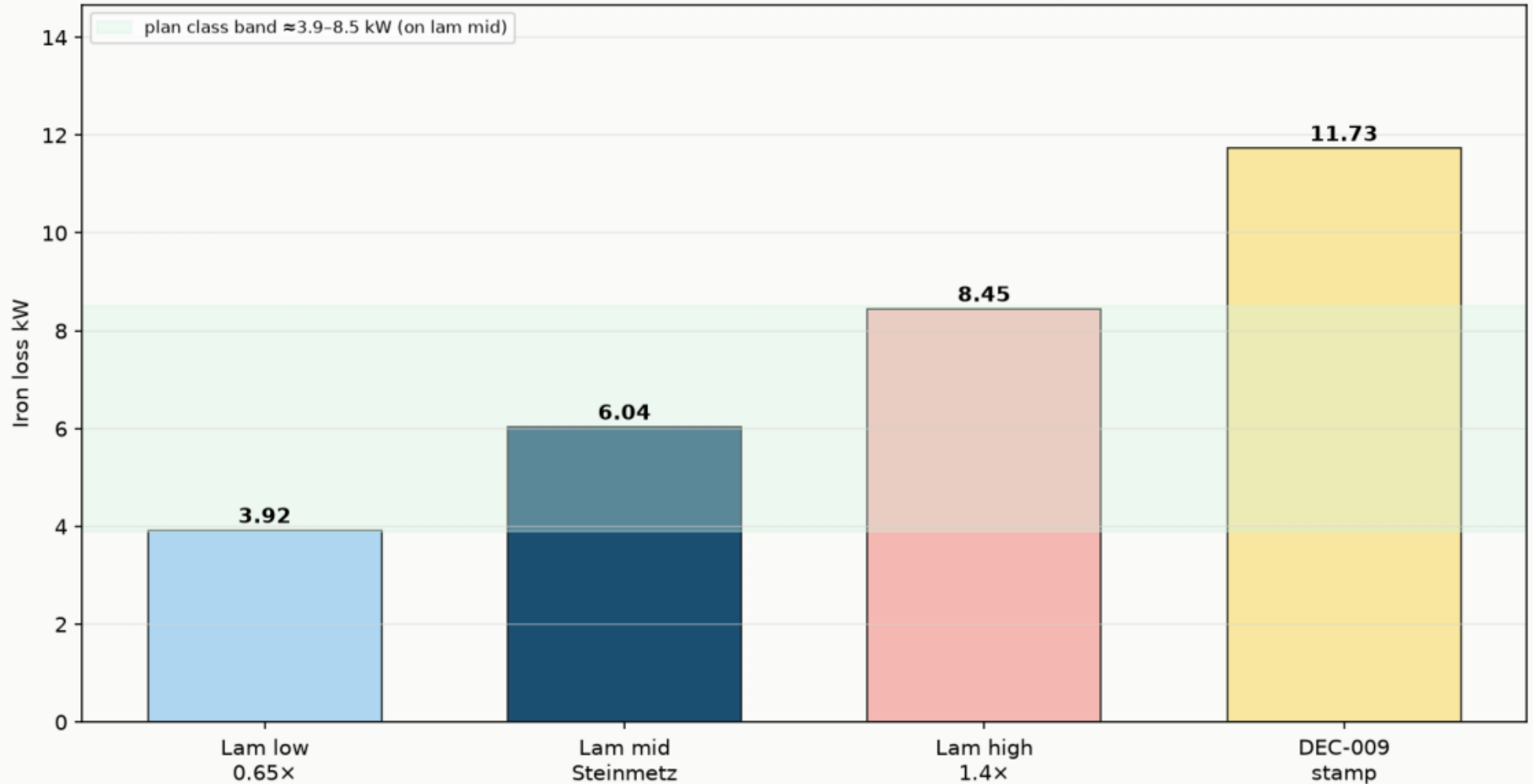
At high 9.90 nH · 8 kA/ μ s: $\Delta V \sim 79$ V · V_{dc} seed 750 V · $I_{dc} \sim 333$ A

HAVE: analytical ESL seeds + target band + cap ESL note

DO NOT HAVE: measured ESL, Eon/Eoff, laminated-bus STEP, SiC MPN (BARB-DOUBLE-PULSE / BARB-SIC-MODULE)

W2.3 · not a double-pulse result · ship_ok=false

Iron loss corners — Jack-facing uncertainty (W2.7)

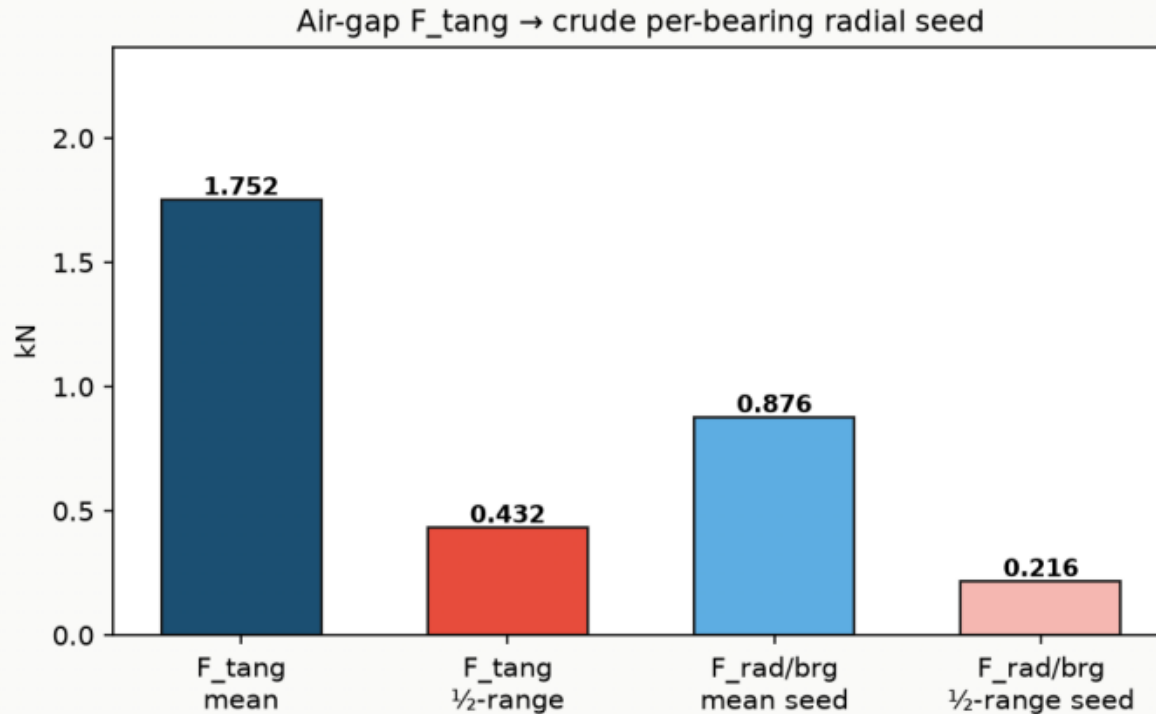


Grade M400-50A · $f_e \approx 1300.0$ Hz · yoke B probe outside pure Steinmetz fit · PWM harmonics OPEN · ship_ok=false · 2026-08-04T14:41:19Z

W2.7 · screening corners only — not a calorimeter result

Bearing reaction — order-of-magnitude under ripple (W2.9)

Upstream ripple status=INPUT_NOT_VALIDATED · L10 OPEN · ship_ok=false · 2026-08-04T14:41:19Z



Assumption (explicit)

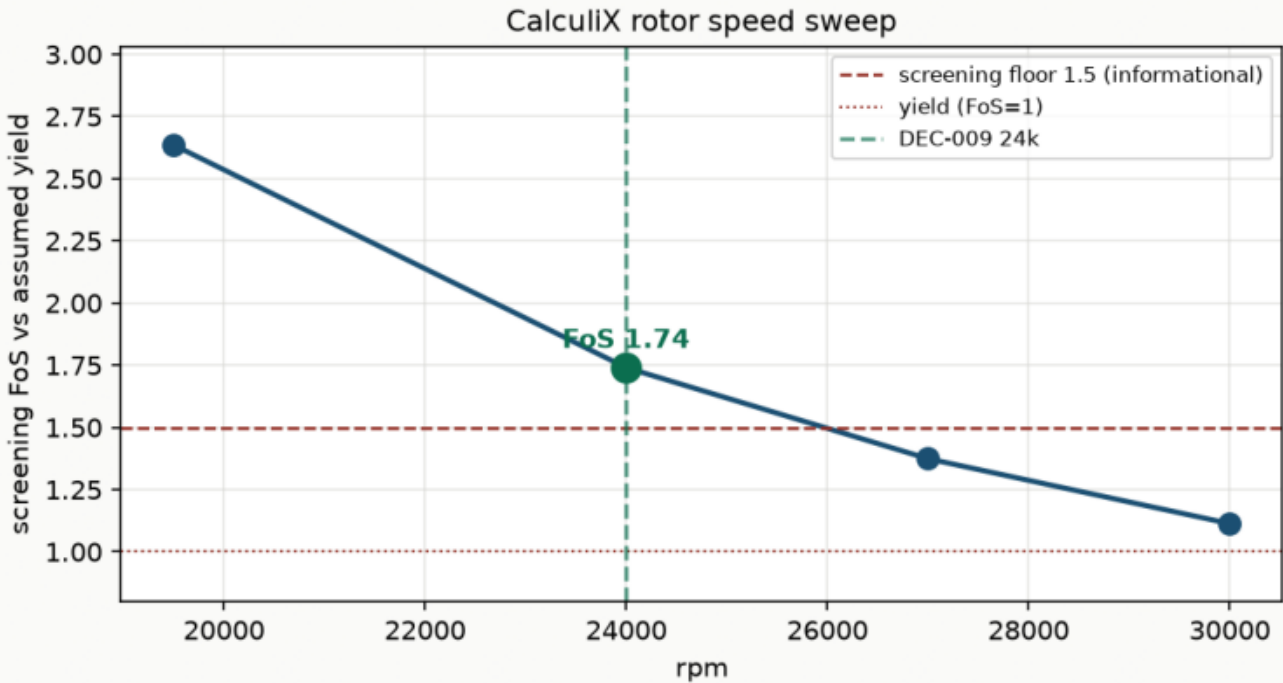
- Two bearings share air-gap tangential force equally
- No gear mesh force
- No magnetic pull unbalance
- No rotor dynamic amplification
- Axial / preload: OPEN
- L10 / ISO 281: OPEN
- Upstream $|T|$ mean 122.09993881083497 N·m

NOT a bearing selection. NOT a life calc. NOT a free-body from CAD mounts.

Partner: BARB-DYNO + shaft/bearing instrumentation before any L10 claim.

Rotor FoS screening card — not release (W2.10)

Assumed yield 355 MPa · release_fos_closed=false · ship_ok=false · 2026-08-04T14:41:19Z



At 24,000 rpm

screening FoS 1.74
 σ_{vm} max 204.079 MPa
yield assumed 355 MPa
below yield True
release closed False

NOT claimed: laminate anisotropy, magnet pocket burst, sleeve retention, fatigue, instrumented overspeed.
Sweep torque_ratio column used legacy machine_torque=81.558081 N·m — not Path B mean.

Partner: BARB-ROTOR-RETENTION (instrumented overspeed to release speed).

W2.10 · screening only · ship_ok=false

NVH / modal — explicit OPEN register (W2.11)

No fake eigenmodes · no invented dB(A) · ship_ok=false · 2026-08-04T14:41:19Z

NVH-001 Housing modal analysis

status=OPEN · blocker: Mount XYZ / stiffness not frozen (B5)

No invented eigenfrequencies

NVH-002 Air-gap force harmonic → structure transfer

status=OPEN · blocker: torque_reliable false; no dyno NVH map

Path B |T| ripple + F_tang OoM exist as excitation seeds only

NVH-003 Gear whine / mesh orders

status=OPEN · blocker: PLANETARY_STRENGTH_VS_ROTOR_BORE; ratio writeback invalidated

NVH-004 Inverter PWM tonal

status=OPEN · blocker: SiC MPN + switching frequency not frozen

NVH-005 Vehicle cabin / chassis transfer

status=OPEN · blocker: Partner vehicle model

Gear-oil screen status — analytical vs B7 CFD (W2.13)

oil_delivery_screen_ok=False · free-surface CFD OPEN · ship_ok=false · 2026-08-04T14:41:19Z

Oil delivery screen

OPEN/FAIL

works_in_kit_context

Cornering pickup

OPEN/FAIL

analytical flag

Tip-speed splash

OK

geometry OoM

Screen numbers (geometry-bound)

carrier speed 2437.5 rpm
churning loss 1202.17 W
gear loss 7.1675 kW
min jet flow 16.4771 L/min
jet pressure need 271.463 kPa
sump volume est. 123.44 ml
Partner: B7 oil bench / free-surface CFD (BARB-GEAR-OIL-BENCH).

CLEARED analytically (under assumptions): tip-speed splash OoM, charge/pickup gallery seeds
WZCFS RESULT UNDER ASSUMPTIONS ship_ok=false
NOT cleared: free-surface CFD, clear-case bench, seal temps, cornering pickup (flag false)

Gear ratio writeback — BLOCKED status (W2.12)

Twin ratio seed = 8 · writeback invalidated · ship_ok=false · 2026-08-04T14:41:19Z

ARCHITECTURE HOLD

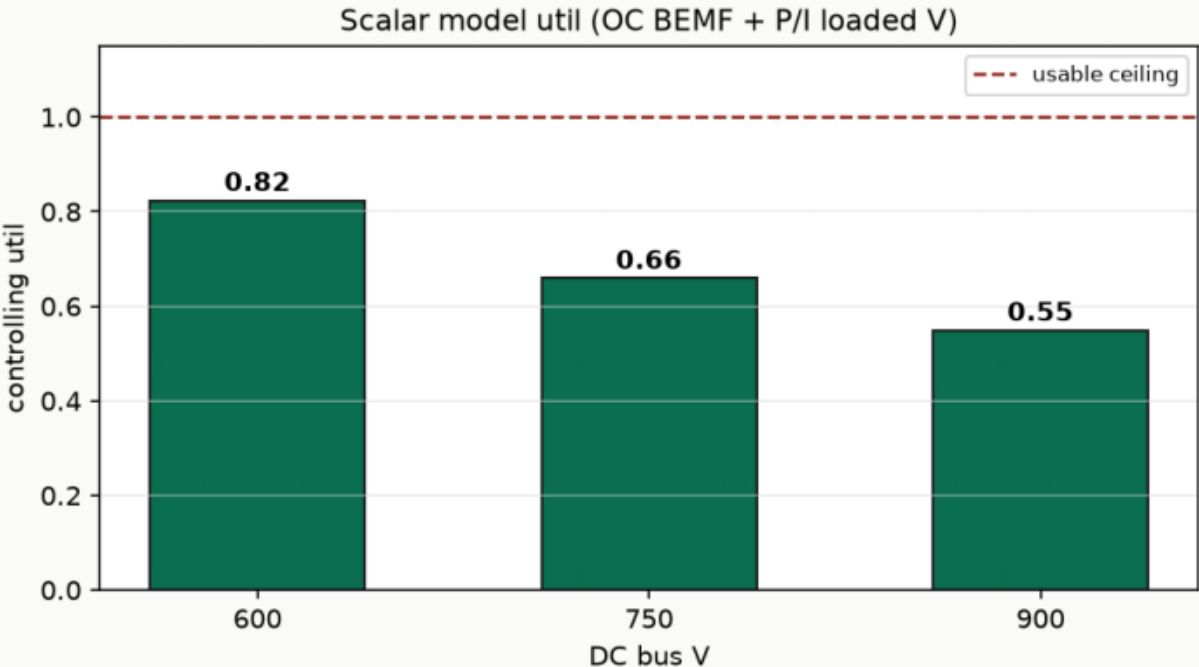
PLANETARY_STRENGTH_VS_ROTOR_BORE – best nest inside rotor ID 105.9 mm reaches FoS ≈ 1.005 (< 1.2); enlarge rotor bore, change tooth counts/topology, or accept external planetary; do not claim PASS

twin gear_ratio	8
writeback.invalidated	True
ISO 6336 duty_strength_ok	False
ISO 6336 min strength FoS	0.1727 ← live screen
hold narrative FoS ≈1.005	(prior nest attempt – NOT same metric)
nest_fits_rotor	True
ratio_matches_twin	True
controlling geometry	packaging_seed_rotor_bore_hold

Do NOT average 1.005 and ISO min FoS – different geometries/attempts.
Unblocks: enlarge rotor bore / retune tooth counts / external planetary,
then re-run gear_geometry_writeback + ISO 6336. Do not invent a PASS ratio nest.
W2.13 gear-oil page is separate (19-gear-oil-status.png).

Path B voltage feasibility — 24k vs 600-900 V (analytical)

speedx1.2308 vs 19.5k · -30° is OP CONTEXT not model input · Bar A open · ship_ok=false · 2026-08-04T15:11:00Z



What this is / is not

- IS:
- Path B OC B_{rms} → BEMF
 - $P/(\eta\sqrt{3} I PF) \rightarrow$ loaded Vll
 - util@750 ≈ 0.659
 - all_fit = True
- IS NOT:
- -30° in voltage vector
 - MTPA voltage circle
 - saturated Ld/Lq map
 - Bar A close
 - torque_reliable

Path B kit-case loaded point is screened near -30° electrical. This voltage screen's back-EMF uses OC B_{rms}; loaded Vll uses $P/(\eta\sqrt{3} I PF)$. Neither equation takes current angle as an input. A true -30° voltage-circle check needs Rs/Ld/Lq (or FEMM flux linkage at load) — OPEN. Do not read this page as MTPA voltage proof.

Prior em_fia_voltage_fw_screen.json = 19.5k / 97.58 mm lineage — not Path B proof.

W4.1-lite · analytical · ship_ok=false · closes_bar_a=false

Dyno request matrix — Path B OP (W5.1 / BARB-DYNO)

Path B mean 122.1 N·m · arch 104.1 · bind 125.2 · 24000 rpm · I_rms 535 A · Vdc 600/750/900 · 60 °C / 12 L/min · ship_ok=false · 2026-08-04T15:11:00Z

ID	Point	rpm	T_nm	I_rms	Vdc	Closes
DYNO-P0	Path B kit OP (primary) Match Path B mean T ≈122.1 N·m; angle context ≈-30° elec (not a locked MTPA)	24000	122.1	535.0	750	torque_reliable / B1-B
DYNO-P1	Architecture duty bar Software architecture bar 104.1 N·m @ 24k / 250 kW class	24000	104.1	measure	750	duty bar correlation
DYNO-P2	Binding ledger bar (stretch) Conservative bind 125.2 N·m — may be thermal/current limited; record limit mode	24000	125.21	measure	750	whether bind is physic
DYNO-P3	Bus min corner Voltage headroom stress at 600 V class	24000	104.1	measure	600	bus-min feasibility /
DYNO-P4	Bus max corner 900 V class — insulation/switching stress	24000	122.1	measure	900	bus-max operating enve
DYNO-P5	Thermal hold @ Path B OP Hold until winding/magnet/module temps stabilise or hit limit; log time history ≥20 Hz	24000	122.1	535.0	750	DEC-008 intermittent v

Required channels

- shaft_torque_nm (calibrated)
- shaft_speed_rpm
- Vdc_v, Idc_a
- phase_I_rms_a (or Ia/Ib/Ic)
- phase_V_ll or Va/Vb/Vc
- coolant_inlet_c, coolant_outlet_c, flow_l_min
- winding_temp_c (or RTD map)
- magnet_temp_proxy_c if available
- module_temp_c
- inverter_faults / derate flags
- timestamp_utc ≥20 Hz preferred; ≥10 Hz minimum

W5.1 · request only — no invented map · torque_reliable stays false until partner returns data · ship_ok=false

Partner asks — ranked (W5 draft)

ship_ok=false · homologation stays ~1/10 · 2026-08-04T15:11:00Z

1. BARB-DYNO	Motor+inverter dyno map @ 60 °C / 12 L/min, 24k class have: Path B 122.1 N·m; matrix page 22 → closes: torque_reliable
2. BARB-DUTY-CYCLE	Lap/stint CSV ≥20 Hz regen power or HV V·I have: DEC-008 24/100 s vignette → closes: DEC-007
3. BARB-FLOW-BENCH	Jacket+plate Δp/flow + wall taps have: Network Δp≈45 kPa; W2.6 → closes: B6
4. BARB-DOUBLE-PULSE	Measured ESL / Eon/Eoff laminated bus have: ESL seeds 4.15–9.9 nH; W2.3 → closes: PE freeze
5. BARB-SIC-MODULE	SiC MPN + datasheet + STEP have: 3-module class packaging → closes: DEC-001
6. BARB-GERBERS	Supplier Gerbers GD + control have: Forge drafts NOT_FAB → closes: B4
7. BARB-ICD-XYZ	Chassis port XYZ mm have: Bay 343×259×267; types ICD → closes: B5
8. BARB-ROTOR-RETENTION	Instrumented overspeed have: CalculiX FoS 1.74 @ 24k screening → closes: B9
9. BARB-GEAR-OIL-BENCH	Clear-case / free-surface oil have: Analytical jet/churning; cornering fail → closes: B7
10. BARB-GEAR-STRENGTH	Unblock PLANETARY_STRENGTH_VS_ROTOR_BORE have: Ratio seed 8; writeback invalidated → closes: W2.12
11. BARB-NVH	Housing modal if mounts frozen have: OPEN register W2.11 (no fake Hz) → closes: NVH
12. BARB-IRON-LOSS	Calorimeter / transient FE iron have: Corner table 3.9–8.5 kW class → closes: Loss

Lap / stint log — required CSV columns (W5.2 / BARB-DUTY-CYCLE)

≥20 Hz preferred · ≥10 Hz minimum · ship_ok=false · 2026-08-04T15:11:00Z

column	unit	need	why
t_s	s	required	Time base, monotonic
vehicle_speed_kph	km/h	optional	Optional cross-check
regen_power_kw	kW	required_or_V_I	Front regen electrical or shaft proxy
Vdc_v	V	required_if_no_power	HV bus
Idc_a	A	required_if_no_power	HV current (signed regen)
front_motor_rpm	rpm	optional	If available
front_torque_nm	N·m	optional	If available
coolant_inlet_c	°C	optional	If logged
throttle_pedal	-	optional	Context only
brake_regen_request	-	optional	Context only

Acceptance

- regen_power_kw OR (Vdc×Idc) continuous
 - no silent constant-fill gaps
 - time base documented
- Closes DFC-007 / R10. Does not invent a duty cycle.

Working hypothesis — provisional partner seeds

Not measured · not ship_ok · replace when partners return · 2026-08-04T15:49:26Z

Rule: design 100% consistently under these seeds. If a seed changes, re-run dependents.

S-EM-TRUTH	Until dyno returns, Path B FE kit-case mean torque is the working EM truth replace → BARB-DYNO measured map @ 24k / 60 °C / 12 L/min
S-DUTY-LAP	Until lap log returns, DEC-008 intermittent vignette is the thermal duty hypot replace → BARB-DUTY-CYCLE lap/stint CSV ≥ 20 Hz
S-COOLANT	Coolant boundary until flow bench replace → BARB-FLOW-BENCH measured Δp /flow/wall taps
S-BUS-PE	DC bus + SiC class + ESL until supplier/double-pulse replace → BARB-SIC-MODULE + BARB-DOUBLE-PULSE + film cap MPN
S-PCB	Forge draft PCBs remain the working topology until supplier Gerbers replace → BARB-GERBERS supplier-stamped stack-up
S-IFACE-XYZ	Bay-local provisional port/mount XYZ for packaging & GA layout replace → BARB-ICD-XYZ team interface control drawing
S-GEAR	Ratio seed 8 held; planetary nest strength remains architecture hold replace → BARB-GEAR-STRENGTH topology decision + BARB-GEAR-OIL-BENCH
S-ROTOR-FOS	Screening FoS 1.74 @ 24k is working structural hypothesis until instrumented o replace → BARB-ROTOR-RETENTION instrumented overspeed
S-NVH	No modal numbers invented; OPEN register is the working state replace → BARB-NVH + frozen mounts
S-IRON-LOSS	Iron loss working band until calorimeter replace → BARB-IRON-LOSS calorimeter / transient FE

Results under those assumptions (Path B bound)

review_status=RESULTS_AVAILABLE_UNDER_ASSUMPTIONS · ship_ok=False · 2026-08-04T15:49:26Z

R-BAY-FIT	Concentric cassette packaging	Housing Ø-xL- mm; rotor ID - / OD - mm
R-EM-DUTY	Electromagnetic duty torque screen (peak	Required at n_max=T/P/ω 104.098914 N·m; FEMM peak 118.8
R-COOL-NET	Coupled cooling network at assumed coola	Δp=45.0826 kPa; T_winding=82.892 °C; T_module=77.61 °C;
R-GEAR-OIL	Geometry-bound gear-oil jet / pickup / c	jet=16.4771 L/min; ΔP_jet=271.463 kPa; churning=1202.17
R-GEAR-PLANET	Planetary strength + nest fit	duty_strength_screen_ok=False; nest_fits_rotor=True
R-BEVEL-DIFF	Bevel differential handbook strength scr	min_strength_FoS=1.2172; contact_FoS=1.2172; duty_stren
R-STRUCT-ROTOR	Rotor ring centrifugal screen @ max rpm	von_Mises=134.7243 MPa; screening_FoS=2.635; below_yiel
R-STRUCT-POCKET	Magnet-pocket iron-bridge centrifugal sc	FEA_von_Mises=190.3724 MPa; FEA_FoS=1.865; analytical_F
R-STRUCT-MOUNT	Cast case / bay-mount torque screen	motor_reaction=125.2193 N·m; carrier_output=1001.7542 N
R-PATH-B-BARS	Dual torque bars under Path B hypothesis	mean=122.099939 N·m; arch=104.098914 N·m; mean/arch=1.1
R-VOLTAGE-24K	Analytical voltage util @ 24k vs 600-900	util@750=0.659; worst_bus=600.0; all_fit=True; closes_b
R-POST-DIFF	Post-diff final-drive software screening	blocker=CLEARED; FoS≥1.2005; bay_fit=True; interfaces_o
R-CAD	Parametric CAD spine (not supplier relea	parametric_family_count=11; release_authority_coverage=

Dense EM map (W4.1 hybrid): 620 pts · peak |T|=106.865174 N·m · torque_map still OPEN · ship_ok false

Gear topology options under provisional seeds

Working hypothesis: OPT-C_EXTERNAL_PLANETARY · ship_ok=false · 2026-08-04T16:06:24Z

OPT-A_NEST_IN_BORE

Nest planetary inside current rotor bore

status=ARCHITECTURE_HOLD bay_fit=True em=none_on_Path_B_geometry

+ Shortest axial stack, No change to Path B EM rotor OD/ID

– PLANETARY_STRENGTH_VS_ROTOR_BORE — best nest inside rotor ID 105.9 mm reaches FoS $\approx$ 1.005

OPT-B_ENLARGE_ROTOR_BORE

Enlarge rotor bore until nest FoS $\geq$ 1.2

status=HYPOTHESIS_CANDIDATE bay_fit=True em=REQUIRES_PATH_B_RERUN

+ Keeps coaxial cassette story, Can clear screening FoS in principle

– Changes rotor structure — Path B EM must be re-solved, Magnet pocket / centrifugal FoS mus

OPT-C_EXTERNAL_PLANETARY

External planetary (outside rotor)

status=HYPOTHESIS_PREFERRED bay_fit=True em=none_Path_B_frozen

+ Rotor bore unconstrained — Path B EM geometry frozen, Strength screen can target FoS $\geq$ 1.2

– Adds gearbox length or width in bay, Extra interfaces (motor-gear coupling)

Freeze Path B EM (122 N·m class) and place gearbox where strength is not hostage to hollow-rotor bore. OPT-C preferred when bay fits; OPT-B only if iron wall remains health

Ratio seed 8 held · Path B EM frozen under OPT-C · not KISSsoft · ship_ok=false

Hypothesis narrative — one page

Full text: FE-FRONT-HYPOTHESIS-NARRATIVE.md

FE Front FPK — design narrative under provisional assumptions

```
**Stamped:** 2026-08-04T16:06:01Z
**Status:** RESULTS_AVAILABLE_UNDER_ASSUMPTIONS
**ship_ok:** false
**Homologation:** ~1/10 until partners replace seeds
```

1. What this document is

Jack is busy. Partners have not returned dyno maps, lap logs, Gerbers, SiC MPNs, chassis XYZ, or bench data. Rather than stall, this twin is frozen under a **named** set of temporary hypotheses. Every number below is either:

1. **Computed** under those hypotheses (screens, FE, packaging), or
2. **An explicit seed** waiting for partner overwrite.

If a seed changes, the dependent results change. That is the contract.

This is **not** race-ready hardware. It is a coherent concept pack under assumptions.

2. The frozen hypotheses (short list)

ID	Working guess	Replaced by
S-EM-TRUTH	Until dyno returns, Path B FE kit-case mean torque is the working EM t	BARB-DYNO measured map @ 24k / 60 °C / 12 L/min
S-DUTY-LAP	Until lap log returns, DEC-008 intermittent vignette is the thermal du	BARB-DUTY-CYCLE lap/stint CSV ≥20 Hz
S-COOLANT	Coolant boundary until flow bench	BARB-FLOW-BENCH measured Δp/flow/wall taps
S-BUS-PE	DC bus + SiC class + ESL until supplier/double-pulse	BARB-SIC-MODULE + BARB-DOUBLE-PULSE + film cap MPN
S-PCB	Forge draft PCBs remain the working topology until supplier Gerbers	BARB-GERBERS supplier-stamped stack-up
S-IFACE-XYZ	Bay-local provisional port/mount XYZ for packaging & GA layout	BARB-ICD-XYZ team interface control drawing
S-GEAR	Ratio seed 8 held; planetary nest strength remains architecture hold	BARB-GEAR-STRENGTH topology decision + BARB-GEAR-0
S-ROTOR-FOS	Screening FoS 1.74 @ 24k is working structural hypothesis until instru	BARB-ROTOR-RETENTION instrumented overspeed
S-NVH	No modal numbers invented; OPEN register is the working state	BARB-NVH + frozen mounts
S-IRON-LOSS	Iron loss working band until calorimeter	BARB-IRON-LOSS calorimeter / transient FE

Full seed file: `\_motor\_stack/provisional\_partner\_seeds.json`

3. Machine we are designing (under S-EM-TRUTH)

- **Topo**

Dense EM campaign — Path B (W4.1)

FEMM dense 11×9 = 99 pts · hybrid map 1488 pts · ship_ok=false · 2026-08-04T16:45:56Z

Dense FEMM grid mode: dense points: 99
Angles °elec: [-20.0, -25.0, -30.0, -35.0, -40.0, -45.0, -50.0, -55.0, -60.0, -65.0, -70.0]
Positions °mech: [0.0, 5.625, 11.25, 16.875, 22.5, 28.125, 33.75, 39.375, 45.0]
I_{rms} design: 535.0 A
Peak |T| dense FEMM: 106.865172 N·m
Best angle (dense): -20.0° elec
Runtime: 2500.4 s
Hybrid total points: 1488
Hybrid peak |T|: 106.865172 N·m

Honesty:

- Path B kit-case position-sweep MEAN remains 122.1 N·m (separate artefact)
- Dense FEMM peak ≠ kit-case mean (different sampling / metric)
- torque_map OPEN · torque_reliable false · ship_ok false